9 Electromagnetic Waves

9.1 Waves in One Dimension

9.1.1 The Wave Equation

What is a "wave"? I don't think I can give you an entirely satisfactory answer – the concept is intrinsically somewhat vague – but here's a start: A wave is a *disturbance of a continuous medium that propagates with a fixed shape at constant velocity*. Immediately I must add qualifiers: in the presence of absorption, the wave will diminish in size as it moves; if the medium is dispersive, different frequencies travel at different speeds; in two or three dimensions, as the wave spreads out, its amplitude will decrease; and of course *standing* waves don't propagate at all. But these are refinements; let's start with the simple case: fixed shape, constant speed, one dimension (Fig. 9.1).

How would you represent such an object mathematically? In Fig. 9.1, I have drawn the wave at two different times, once at $t = 0$, and again at some later time t – each point on the wave form simply shifts to the right by an amount vt, where v is the velocity. Maybe the wave is generated by shaking one end of a taut string; $f(z, t)$ represents the displacement of the string at the point z, at time t. Given the *initial* shape of the string, $g(z) \equiv f(z, 0)$, what is the subsequent form, $f(z, t)$? Well, the displacement at point z, at the later time t, is the same as the displacement a distance vt to the left (that is, at $z - vt$), back at time $t = 0$:

$$f(z, t) = f(z - vt, 0) = g(z - vt). \tag{9.1}$$

That statement captures (mathematically) the essence of wave motion. It tells us that the function $f(z, t)$, which *might* have depended on z and t in *any* old way, in *fact* depends on them only in the very special combination $z - vt$; when that is true, the function $f(z, t)$ represents a wave of fixed shape traveling in the z direction at speed v. For example, if A and b are constants (with the appropriate units),

$$f_1(z, t) = Ae^{-b(z-vt)^2}, \quad f_2(z, t) = A\sin[b(z - vt)], \quad f_3(z, t) = \frac{A}{b(z - vt)^2 + 1}$$

all represent waves (with different shapes, of course), but

$$f_4(z, t) = Ae^{-b(bz^2+vt)} \quad \text{and} \quad f_5(z, t) = A\sin(bz)\cos(bvt)^3$$

do *not*.

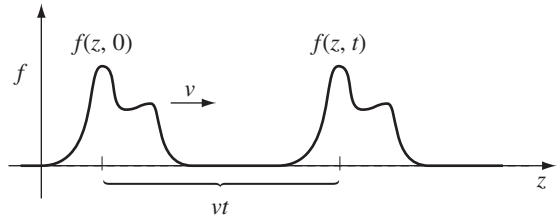


Fig. 9.1

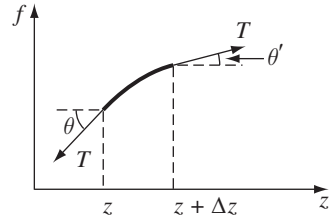


Fig. 9.2

Why does a stretched string support wave motion? Actually, it follows from Newton's second law. Imagine a very long string under tension T. If it is displaced from equilibrium, the net transverse force on the segment between z and $z + \Delta z$ (Fig. 9.2) is

$$\Delta F = T \sin \theta' - T \sin \theta,$$

where θ' is the angle the string makes with the z direction at point $z + \Delta z$, and θ is the corresponding angle at point z. Provided that the distortion of the string is not too great, these angles are small (the figure is exaggerated, obviously), and we can replace the sine by the tangent:

$$\Delta F \approx T(\tan \theta' - \tan \theta) = T\Big(\frac{\partial f}{\partial z}\Big|_{z+\Delta z} - \frac{\partial f}{\partial z}\Big|_{z}\Big) \approx T\frac{\partial^2 f}{\partial z^2}\Delta z.$$

If the mass per unit length is μ, Newton's second law says

$$\Delta F = \mu(\Delta z)\frac{\partial^2 f}{\partial t^2},$$

and therefore

$$\frac{\partial^2 f}{\partial z^2} = \frac{\mu}{T}\frac{\partial^2 f}{\partial t^2}.$$

Evidently, small disturbances on the string satisfy

$$\boxed{\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2},} \tag{9.2}$$

where v (which, as we'll soon see, represents the speed of propagation) is

$$v = \sqrt{\frac{T}{\mu}}. \tag{9.3}$$

Equation 9.2 is known as the (classical) **wave equation**, because it admits as solutions all functions of the form

$$f(z,t) = g(z - vt) \tag{9.4}$$

(that is, all functions that depend on the variables z and t in the special combination $u \equiv z - vt$), and we have just learned that such functions represent waves propagating in the z direction with speed v.

Proof.

$$\frac{\partial f}{\partial z} = \frac{dg}{du}\frac{\partial u}{\partial z} = \frac{dg}{du}, \quad \frac{\partial f}{\partial t} = \frac{dg}{du}\frac{\partial u}{\partial t} = -v\frac{dg}{du},$$

and

$$\frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z}\left(\frac{dg}{du}\right) = \frac{d^2g}{du^2}\frac{\partial u}{\partial z} = \frac{d^2g}{du^2},$$

$$\frac{\partial^2 f}{\partial t^2} = -v\frac{\partial}{\partial t}\left(\frac{dg}{du}\right) = -v\frac{d^2g}{du^2}\frac{\partial u}{\partial t} = v^2\frac{d^2g}{du^2},$$

so

$$\frac{d^2g}{du^2} = \frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}. \qquad \square$$

Note that $g(u)$ can be *any* (differentiable) *function whatever*.[1] If the disturbance propagates at constant speed without changing its shape, then it satisfies the wave equation.

But functions of the form $g(z - vt)$ are not the *only* solutions. The wave equation involves the *square* of v, so we can generate another class of solutions by simply changing the sign of the velocity:

$$f(z,t) = h(z + vt). \tag{9.5}$$

This, of course, represents a wave propagating in the *negative* z direction, and it is certainly reasonable (on physical grounds) that such solutions would be allowed. What *is* perhaps surprising is that the *most general* solution to the wave equation is the sum of a wave to the right and a wave to the left:

$$f(z,t) = g(z - vt) + h(z + vt). \tag{9.6}$$

(Notice that the wave equation is **linear**: the sum of any two solutions is itself a solution.) As it turns out, *every* solution to the wave equation can be expressed in this form.

Like the simple harmonic oscillator equation, the wave equation is ubiquitous in physics. If something is vibrating, the oscillator equation is almost certainly responsible (at least, for small amplitudes), and if something is waving (whether the context is mechanics or acoustics, optics

[1] Does it surprise you that the class of solutions to the wave equation is so huge? After all, an *ordinary* second-order differential equation typically admits just *two* linearly independent solutions …. That's right, but there's no such rule for *partial* differential equations.

or oceanography), the wave equation (perhaps with some decoration) is bound to be involved.

Problem 9.1. By explicit differentiation, check that the functions f_1, f_2, and f_3 in the text satisfy the wave equation. Show that f_4 and f_5 do *not*.

Problem 9.2. Show that the **standing wave** $f(z,t) = A\sin(kz)\cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right (Eq. 9.6).

9.1.2 Sinusoidal Waves

(i) Terminology. Of all possible wave forms, the sinusoidal one

$$f(z,t) = A\cos[k(z - vt) + \delta] \tag{9.7}$$

is (for good reason) the most familiar. Figure 9.3 shows this function at time $t = 0$. The quantity A is the **amplitude** of the wave (it is positive, and represents the maximum displacement from equilibrium). The argument of the cosine is called the **phase**, and δ is the **phase constant** (obviously, you can add any integer multiple of 2π to δ without changing $f(z,t)$; ordinarily, one uses a value in the range $0 \le \delta < 2\pi$, but sometimes $-\pi < \delta \le \pi$, for example, may be more convenient). Notice that at $z = vt - \delta/k$, the phase is zero; let's call this the "central maximum." If $\delta = 0$, the central maximum passes the origin at time $t = 0$; more generally, δ/k is the distance by which the central maximum (and therefore the entire wave) is "delayed." Finally, k is the **wave number**; it is related to the **wavelength** λ by the equation

$$\lambda = \frac{2\pi}{k}, \tag{9.8}$$

for when z advances by $2\pi/k$, the cosine executes one complete cycle.

As time passes, the entire wave train proceeds to the right, at speed v. At any given point z, the string vibrates up and down, undergoing one full cycle in a **period**

$$T = \frac{2\pi}{kv}. \tag{9.9}$$

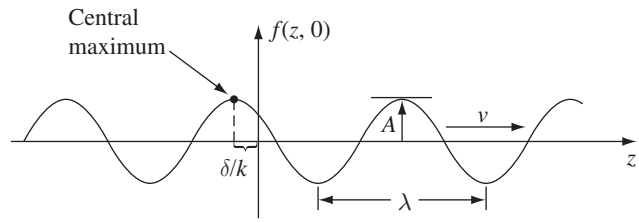


Fig. 9.3

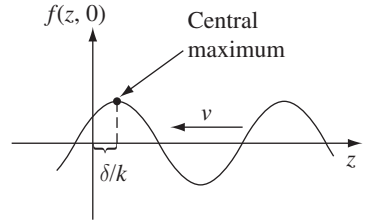


Fig. 9.4

The **frequency** ν (number of oscillations per unit time) is

$$\nu = \frac{1}{T} = \frac{kv}{2\pi} = \frac{v}{\lambda}. \tag{9.10}$$

For our purposes, a more convenient unit is the **angular frequency** ω, so called because in the analogous case of uniform circular motion, it represents the number of radians swept out per unit time:

$$\omega = 2\pi\nu = kv, \quad \text{so} \quad \boxed{\omega = kv.} \tag{9.11}$$

Ordinarily, it's nicer to write sinusoidal waves (Eq. 9.7) in terms of ω, instead of v:[2]

$$f(z,t) = A\cos(kz - \omega t + \delta). \tag{9.12}$$

A sinusoidal oscillation of wave number k and (angular) frequency ω traveling to the *left* would be written

$$f(z,t) = A\cos(kz + \omega t - \delta). \tag{9.13}$$

The sign of the phase constant is chosen for consistency with our previous convention that δ/k shall represent the distance by which the wave is "delayed" (since the wave is now moving to the *left*, a delay means a shift to the *right*). At $t = 0$, the wave looks like Fig. 9.4. Because the cosine is an *even* function, we can just as well write Eq. 9.13 thus:

$$f(z,t) = A\cos(-kz - \omega t + \delta). \tag{9.14}$$

Comparison with Eq. 9.12 reveals that *we could simply switch the sign of k* to produce a wave with the same amplitude, phase constant, frequency, and wavelength, traveling in the opposite direction.

(ii) Complex notation. In view of **Euler's formula**,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{9.15}$$

the sinusoidal wave (Eq. 9.12) can be written

$$f(z,t) = \mathrm{Re}[Ae^{i(kz-\omega t+\delta)}], \tag{9.16}$$

[2] I put a box around the equation $\omega = kv$, because the relation between wave number (or wavelength) and frequency (ν or ω) is fundamental, and it really *ought* to have a memorable name. Essentially the same idea can be expressed in several different ways: $\nu\lambda = v$, $\lambda\omega = 2\pi v$, etc.

where $\mathrm{Re}(\xi)$ denotes the real part of the complex number ξ. This invites us to introduce the **complex wave function**

$$\tilde{f}(z,t) \equiv \tilde{A}e^{i(kz-\omega t)}, \tag{9.17}$$

with the **complex amplitude** $\tilde{A} \equiv Ae^{i\delta}$ subsuming both the (positive) real amplitude A and the phase constant δ. The *actual* wave function is the real part of $\tilde{f}$:

$$f(z,t) = \mathrm{Re}[\tilde{f}(z,t)]. \tag{9.18}$$

If you know $\tilde{f}$, it is a simple matter to find f; the *advantage* of the complex notation is that exponentials are much easier to manipulate than sines and cosines.

Example 9.1. Suppose you want to add two sinusoidal waves:

$$f_3 = f_1 + f_2 = \mathrm{Re}(\tilde{f}_1) + \mathrm{Re}(\tilde{f}_2) = \mathrm{Re}(\tilde{f}_1 + \tilde{f}_2) = \mathrm{Re}(\tilde{f}_3),$$

with $\tilde{f}_3 = \tilde{f}_1 + \tilde{f}_2$. You simply add the corresponding *complex* wave functions, and then take the real part. In particular, if they have the same frequency and wave number,

$$\tilde{f}_3 = \tilde{A}_1 e^{i(kz-\omega t)} + \tilde{A}_2 e^{i(kz-\omega t)} = \tilde{A}_3 e^{i(kz-\omega t)},$$

where

$$\tilde{A}_3 = \tilde{A}_1 + \tilde{A}_2, \quad \text{or } A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}. \tag{9.19}$$

In other words, you just add the (complex) amplitudes. The combined wave still has the same frequency and wavelength,

$$f_3(z,t) = A_3 \cos(k_Z - \omega t + \delta_3),$$

and you can easily figure out A_3 and δ_3 from Eq. 9.19 (see Prob. 9.3). Try doing this *without* using the complex notation – you will find yourself looking up trig identities and wading through nasty algebra.

Caveat: The complex notation is great for *adding* sinusoidal functions, because

$$\mathrm{Re}(\tilde{f}_1 + \tilde{f}_2) = \mathrm{Re}(\tilde{f}_1) + \mathrm{Re}(\tilde{f}_2),$$

but not so useful for *multiplying* them:

$$\mathrm{Re}(\tilde{f}_1 \cdot \tilde{f}_2) \neq \mathrm{Re}(\tilde{f}_1) \cdot \mathrm{Re}(\tilde{f}_2).$$

Be on guard, therefore, when you encounter quantities (energy, for example) that are quadratic. There are tricks for handling special cases (see Prob. 9.12), but the fireproof method is to revert to the real notation.

(iii) Linear combinations of sinusoidal waves. Although the sinusoidal function (Eq. 9.17) is a very special wave form, the fact is that *any* wave can be expressed as a linear combination of sinusoidal ones:

$$\tilde{f}(z,t) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i(kz-\omega t)}\, dk. \tag{9.20}$$

Here ω is a function of k (Eq. 9.11), and I have allowed k to run negative in order to include waves going both directions.[3]

The formula for $\tilde{A}(k)$, in terms of the initial conditions $f(z,0)$ and $\dot{f}(z,0)$, can be obtained from the theory of Fourier transforms (see Prob. 9.34), but the details are not relevant to my purpose here. The point is that *any* wave can be written as a linear combination of sinusoidal waves, and therefore if you know how sinusoidal waves behave, you know in principle how *any* wave behaves. So from now on, we shall confine our attention to sinusoidal waves.

Problem 9.3. Use Eq. 9.19 to determine A_3 and δ_3 in terms of A_1, A_2, δ_1, and δ_2.

Problem 9.4. Obtain Eq. 9.20 directly from the wave equation, by separation of variables.

9.1.3 Boundary Conditions: Reflection and Transmission

So far, I have assumed the string is infinitely long – or at any rate long enough that we don't need to worry about what happens to a wave when it reaches the end. As a matter of fact, what happens depends a lot on how the string is *attached – that is*, on the specific boundary conditions to which the wave is subject. Suppose, for instance, that the string is simply tied onto a *second* string. The tension T is the same for both, but the mass per unit length μ, presumably, is not, and hence the wave velocities v_1 and v_2 are different (remember, $v = \sqrt{T/\mu}$). Let's say, for convenience, that the knot occurs at $z = 0$. The **incident** wave

$$\tilde{f}_I(z,t) = \tilde{A}_I e^{i(k_1 z-\omega t)} \quad (z < 0), \tag{9.21}$$

coming in from the left, gives rise to a **reflected** wave

$$\tilde{f}_R(z,t) = \tilde{A}_R e^{i(-k_1 z-\omega t)} \quad (z < 0), \tag{9.22}$$

traveling *back* along string 1 (hence the minus sign in front of k_1), in addition to a **transmitted** wave

$$\tilde{f}_T(z,t) = \tilde{A}_T e^{i(k_2 z-\omega t)} \quad (z > 0), \tag{9.23}$$

which continues on to the right in string 2.

[3] This does not mean that λ and ω are negative – wavelength and frequency are *intrinsically* positive (like speed v and amplitude A). If we allow negative wave numbers, then Eqs. 9.8 and 9.11 should really be written $\lambda = 2\pi/|k|$ and $\omega = |k|v$.

The incident wave $f_I(z, t)$ is a sinusoidal oscillation that extends (in principle) all the way back to $z = -\infty$, and has been doing so for all of history. The same goes for f_R and f_T (except that the latter, of course, extends to $z = +\infty$). *All parts of the system are oscillating at the same frequency* ω (a frequency determined by the person at $z = -\infty$, who is shaking the string in the first place). Since the wave velocities are different in the two strings, however, the wavelengths and wave numbers are also different:

$$\frac{\lambda_1}{\lambda_2} = \frac{k_2}{k_1} = \frac{v_1}{v_2}. \tag{9.24}$$

Of course, this situation is pretty artificial – what's more, with incident and reflected waves of infinite extent traveling on the same piece of string, it's going to be hard for a spectator to tell them apart. You might therefore prefer to consider an incident wave of *finite* extent – say, the pulse shown in Fig. 9.5. You can work out the details for yourself, if you like (Prob. 9.5). The trouble with this approach is that no *finite* pulse is truly sinusoidal. The waves in Fig. 9.5 may *look* like sine functions, but they're *not*: they're little *pieces* of sines, joined onto an entirely different function (namely, zero). Like any other waves, they can be built up as *linear combinations* of true sinusoidal functions (Eq. 9.20), but only by putting together a whole range of frequencies and wavelengths. If you want a *single* incident frequency (as we shall in the electromagnetic case), you must let your waves extend to infinity. (In practice, if you use a very long pulse, with many oscillations, it will be *close* to the ideal of a single frequency.)

For a sinusoidal incident wave, then, the net disturbance of the string is

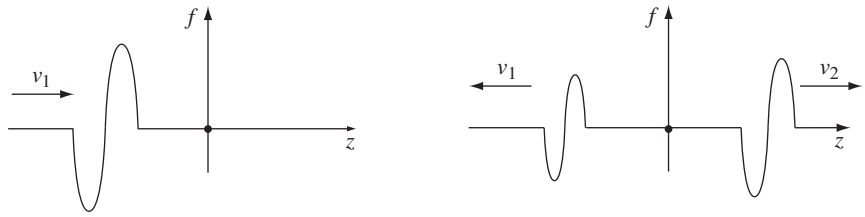
$$\tilde{f}(z, t) = \begin{cases} \tilde{A}_I e^{i(k_1 z - \omega t)} + \tilde{A}_R e^{i(-k_1 z - \omega t)} & \text{for } z < 0, \\ \tilde{A}_T e^{i(k_2 z - \omega t)} & \text{for } z > 0. \end{cases} \tag{9.25}$$

At the join ($z = 0$), the displacement just slightly to the left ($z = 0^-$) must equal the displacement slightly to the right ($z = 0^+$) – else there would be a break in the string. Mathematically, $f(z, t)$ is *continuous* at $z = 0$:

$$f(0^-, t) = f(0^+, t). \tag{9.26}$$

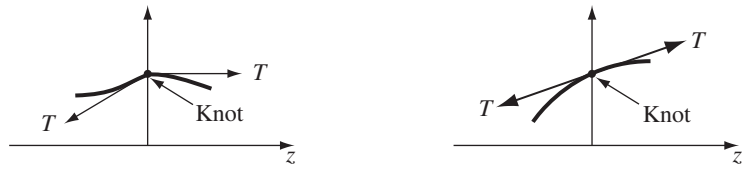
If the knot itself is of negligible mass, then the *derivative* of f must *also* be continuous:

$$\frac{\partial f}{\partial z}\Big|_{0^-} = \frac{\partial f}{\partial z}\Big|_{0^+}. \tag{9.27}$$



(a) Incident pulse (b) Reflected and transmitted pulses

Fig. 9.5



(a) Discontinuous slope; force on knot (b) Continuous slope; no force on knot

Fig. 9.6

Otherwise there would be a net force on the knot, and therefore an infinite acceleration (Fig. 9.6). These boundary conditions apply directly to the *real* wave function $f(z,t)$. But since the imaginary part of $\tilde{f}$ differs from the real part only in the replacement of cosine by sine (Eq. 9.15), it follows that the complex wave function $\tilde{f}(z,t)$ obeys the same rules:

$$\tilde{f}(0^-,t) = \tilde{f}(0^+,t), \quad \frac{\partial \tilde{f}}{\partial z}\Big|_{0^-} = \frac{\partial \tilde{f}}{\partial z}\Big|_{0^+}. \tag{9.28}$$

When applied to Eq. 9.25, these boundary conditions determine the outgoing amplitudes ($\tilde{A}_R$ and $\tilde{A}_T$) in terms of the incoming one ($\tilde{A}_I$):

$$\tilde{A}_I + \tilde{A}_R = \tilde{A}_T, \quad k_1(\tilde{A}_I - \tilde{A}_R) = k_2\tilde{A}_T,$$

from which it follows that

$$\tilde{A}_R = \Big(\frac{k_1 - k_2}{k_1 + k_2}\Big)\tilde{A}_I, \quad \tilde{A}_T = \Big(\frac{2k_1}{k_1 + k_2}\Big)\tilde{A}_I. \tag{9.29}$$

Or, in terms of the velocities (Eq. 9.24):

$$\tilde{A}_R = \Big(\frac{v_2 - v_1}{v_2 + v_1}\Big)\tilde{A}_I, \quad \tilde{A}_T = \Big(\frac{2v_2}{v_2 + v_1}\Big)\tilde{A}_I. \tag{9.30}$$

The *real* amplitudes and phases, then, are related by

$$A_R e^{i\delta_R} = \Big(\frac{v_2 - v_1}{v_2 + v_1}\Big)A_I e^{i\delta_I}, \quad A_T e^{i\delta_T} = \Big(\frac{2v_2}{v_2 + v_1}\Big)A_I e^{i\delta_I}. \tag{9.31}$$

If the second string is *lighter* than the first ($\mu_2 < \mu_1$, so that $v_2 > v_1$), all three waves have the same phase angle ($\delta_R = \delta_T = \delta_I$), and the outgoing amplitudes are

$$A_R = \Big(\frac{v_2 - v_1}{v_2 + v_1}\Big)A_I, \quad A_T = \Big(\frac{2v_2}{v_2 + v_1}\Big)A_I. \tag{9.32}$$

If the second string is *heavier* than the first ($v_2 < v_1$), the reflected wave is out of phase by 180° ($\delta_R + \pi = \delta_T = \delta_I$). In other words, since

$$\cos(-k_1 z - \omega t + \delta_I - \pi) = -\cos(-k_1 z - \omega t + \delta_I),$$

the reflected wave is "upside down." The amplitudes in this case are

$$A_R = \Big(\frac{v_1 - v_2}{v_2 + v_1}\Big)A_I, \quad A_T = \Big(\frac{2v_2}{v_2 + v_1}\Big)A_I. \tag{9.33}$$

In particular, if the second string is *infinitely* massive – or, what amounts to the same thing, if the first string is simply tied to an immovable post at the end – then

$$A_R = A_I \quad \text{and} \quad A_T = 0.$$

Naturally, in this case there is *no* transmitted wave – *all* of it reflects back.

! **Problem 9.5.** Suppose you send an incident wave of specified shape, $g_I(z - v_1 t)$, down string 1. It gives rise to a reflected wave, $h_R(z + v_1 t)$, and a transmitted wave, $g_T(z - v_2 t)$. By imposing the boundary conditions 9.26 and 9.27, find h_R and g_T.

Problem 9.6.

(a) Formulate an appropriate boundary condition, to replace Eq. 9.27, for the case of two strings under tension T joined by a knot of mass m.
(b) Find the amplitude and phase of the reflected and transmitted waves for the case where the knot has mass m and the second string is massless.

! **Problem 9.7.** Suppose string 2 is embedded in a viscous medium (such as molasses), which imposes a drag force that is proportional to its (transverse) speed:

$$\Delta F_{\text{drag}} = -\gamma \frac{\partial f}{\partial t} \Delta z.$$

(a) Derive the modified wave equation describing the motion of the string.
(b) Solve this equation, assuming the string vibrates at the incident frequency ω. That is, look for solutions of the form $\tilde{f}(z, t) = e^{-i\omega t}\tilde{F}(z)$.
(c) Show that the waves are **attenuated** (that is, their amplitude decreases with increasing z). Find the characteristic penetration distance, at which the amplitude is $1/e$ of its original value, in terms of γ, T, μ, and ω.
(d) If a wave of amplitude A_I, phase $\delta_I = 0$, and frequency ω is incident from the left (string 1), find the reflected wave's amplitude and phase.

9.1.4 Polarization

The waves that travel down a string when you shake it are called **transverse**, because the displacement is perpendicular to the direction of propagation. If the string is reasonably elastic, it is also possible to stimulate *compression* waves, by giving the string little tugs. Compression waves are hard to see on a string, but if you try it with a slinky they're quite noticeable (Fig. 9.7). These waves are called **longitudinal**, because the displacement from equilibrium is along the direction of propagation. Sound waves, which are nothing but compression waves in air, are longitudinal; electromagnetic waves, as we shall see, are transverse.

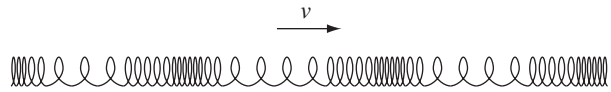
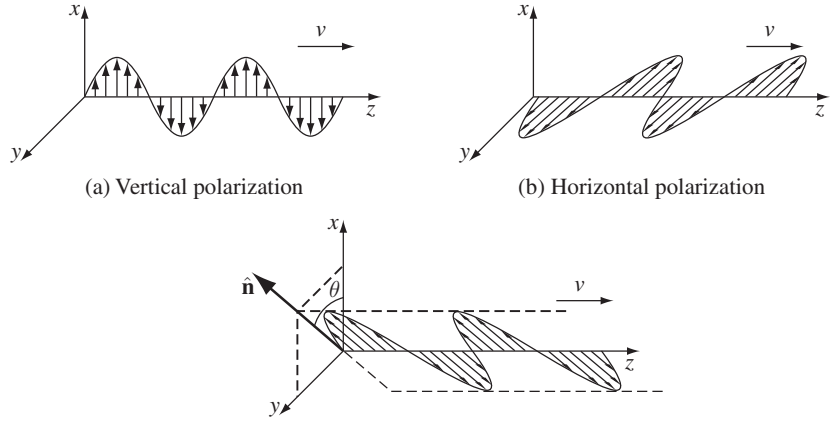


Fig. 9.7



(a) Vertical polarization (b) Horizontal polarization

(c) Polarization vector

Fig. 9.8

Now there are, of course, *two* dimensions perpendicular to any given line of propagation. Accordingly, transverse waves occur in two independent states of **polarization**: you can shake the string up-and-down ("vertical" polarization – Fig. 9.8a),

$$\tilde{\mathbf{f}}_v(z,t) = \tilde{A}e^{i(kz-\omega t)}\hat{\mathbf{x}}, \tag{9.34}$$

or left-and-right ("horizontal" polarization – Fig. 9.8b),

$$\tilde{\mathbf{f}}_h(z,t) = \tilde{A}e^{i(kz-\omega t)}\hat{\mathbf{y}}, \tag{9.35}$$

or along any other direction in the xy plane (Fig. 9.8c):

$$\tilde{\mathbf{f}}(z,t) = \tilde{A}e^{i(kz-\omega t)}\hat{\mathbf{n}}. \tag{9.36}$$

The **polarization vector** $\hat{\mathbf{n}}$ defines the plane of vibration.[4] Because the waves are transverse, $\hat{\mathbf{n}}$ is perpendicular to the direction of propagation:

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{z}} = 0. \tag{9.37}$$

In terms of the **polarization angle** θ,

$$\hat{\mathbf{n}} = \cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}}. \tag{9.38}$$

Thus, the wave pictured in Fig. 9.8c can be considered a superposition of two waves – one horizontally polarized, the other vertically:

$$\tilde{\mathbf{f}}(z,t) = (\tilde{A}\cos\theta)e^{i(kz-\omega t)}\hat{\mathbf{x}} + (\tilde{A}\sin\theta)e^{i(kz-\omega t)}\hat{\mathbf{y}}. \tag{9.39}$$

[4] Notice that you can always switch the *sign* of $\hat{\mathbf{n}}$, provided you simultaneously advance the phase constant by 180°, since both operations change the sign of the wave function.